can be used for automating tests and generating test reports.

Visual Basic. It is a very old programming language used by engineers. With some knowledge and experience, one can easily automate the tests.

Normally, software architecture has three main steps:

Initialisation. In this, all logic for power-up and configuration are done

Testing. In this, all sequences of the test can be done

Clean up. In this, all shutdown logic can be implemented

Test reports

The test report is the most important part of the test. Reports should be easy to read and transfer to concerned people.

Normally used format is .csv or xls. The advantage is that it is very easy to create, and most software can read it like Excel and Open Office. The disadvantage is that format may change while importing the files. This can be solved by some delimiters, but for large data, it is a very difficult and time-consuming task.

Another format is HTML, which is useful in making a colourful report. Pass and fail tests can be easily marked for the user. Also, HTML files can be viewed on a normal browser, which is a great advantage for some.

PDF is also one of the most popular formats used for test reports. Making reports in PDF may require some plug-ins, which depends upon the test software.

Data storage

Storing the data is the last stage of the test, which is not taken care of by many users. Test data can be stored directly in some folders or a database.

The advantage of storing test data in a database is that it cannot be easily deleted, and only an authorised person can access it. One more advantage of storing the data in a database is that one can analyse the stored test data for debugging.

Cost

Finally, whether to go for automated testing or not depends on the number of DUT to be tested in a day or month. For example, if you have more than thirty DUT to test in a day, it is a good criterion to go for automated tests.

The cost of an automated test jig depends on many factors like the number of test points, the complexity of the test, and the size of DUT. The cost can be divided into mechanical structure, DAQ board design, and software design. Normally, for a board with thirty test points, the cost can go up to ₹ 50,000.

Time and training for users

The time required for testing thirty test points test jigs would not be more than five minutes. However, complex tests require more time.

Extensive training is not required as the user will only monitor the test. In some cases, basic training for DUT and automation software application usage is required.

Conclusion

More and more electronic industries are shifting to automated tests from manual tests. This saves time and resources like multimetres, function generators, and oscilloscopes. Design teams can focus only on the design part. Some EMS vendors also support test jigs. EFY